

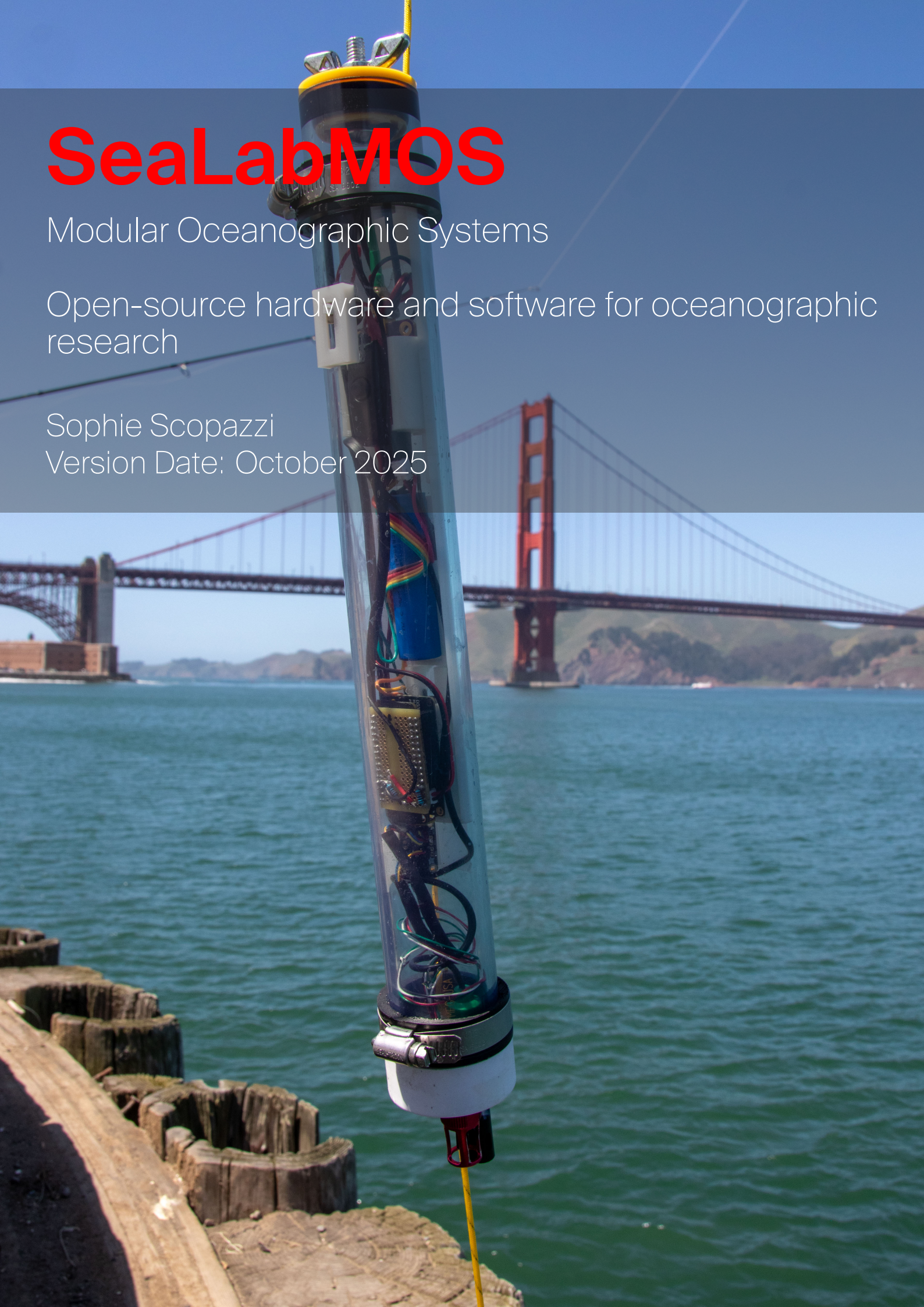
SeaLabMOS

Modular Oceanographic Systems

Open-source hardware and software for oceanographic research

Sophie Scopazzi

Version Date: October 2025



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Preface

This manual remains a work in progress. Please look at version date below.

If you have experience with oceanographic sampling systems and methods, I recommend skipping to chapter 2.

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1

Ocean Sampling Background

Chapter Concepts:

1. What is a CTD and what are they used for?
2. How are research platforms typically deployed in the ocean?
3. What sensor platforms are adjacent to a CTD?

1.1. What is a CTD?

CTD (for **C**onductivity, **T**emperature, and **D**epth) refers to the three primary parameters measured by the instrument. Salinity is not measured directly but calculated from conductivity, temperature, and pressure, since the salinity of water affects its electrical conductivity. Data from a CTD can be used to plot salinity and temperature profiles as a function of depth. An example of what this can look like is on the next page, Fig. 1.1.

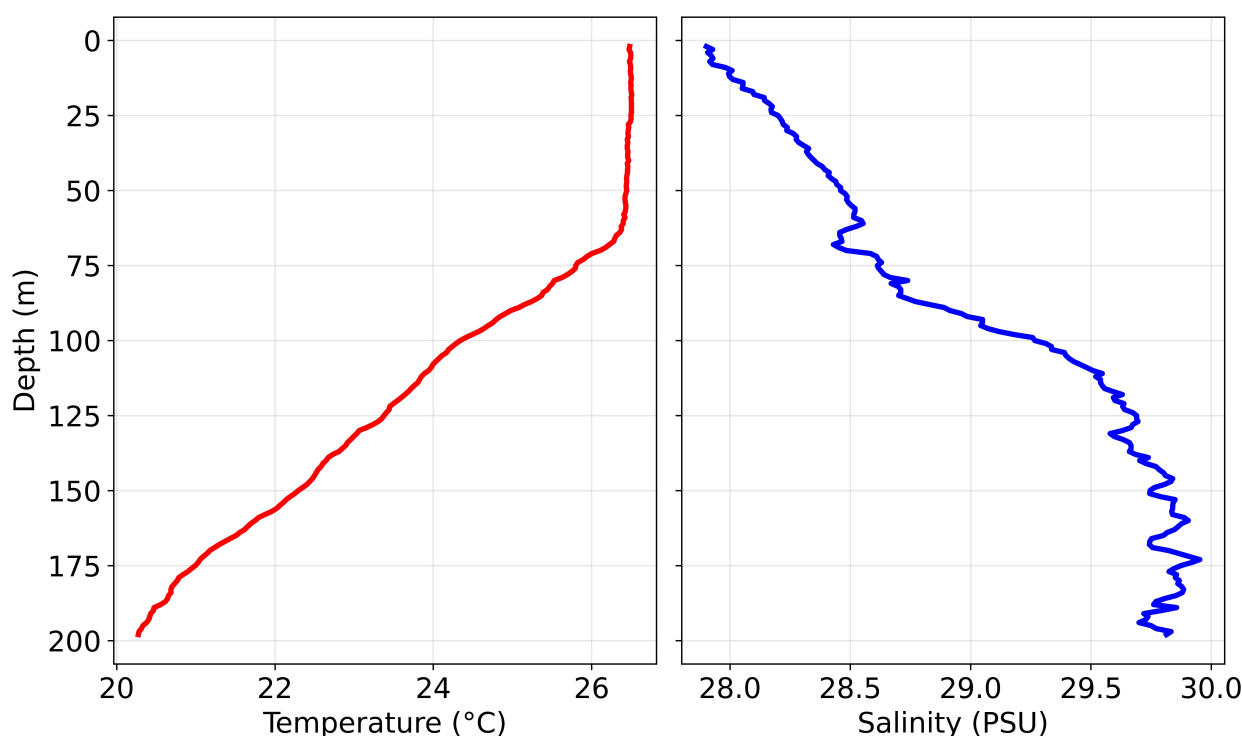


Figure 1.1: 200 meter cast of the two-inch steel SeaLabMOS CTD

Another common method to display CTD data is called a TS diagram (short for temperature-salinity) because the plot compares the two variables, typically using color to indicate depth. Fig. 1.2 shows that the surface water was warmer and less salty, with temperature decreasing and salinity increasing with depth.

1.2. CTD Applications

CTD data can be used to answer questions such as whether the water column is stable and resistant to vertical mixing or unstable and prone to mixing, and to determine the depth of the thermocline or halocline. This information also assists with further study of the micro-organisms in the epipelagic zone, the upper photic layer of the ocean where sunlight penetrates.

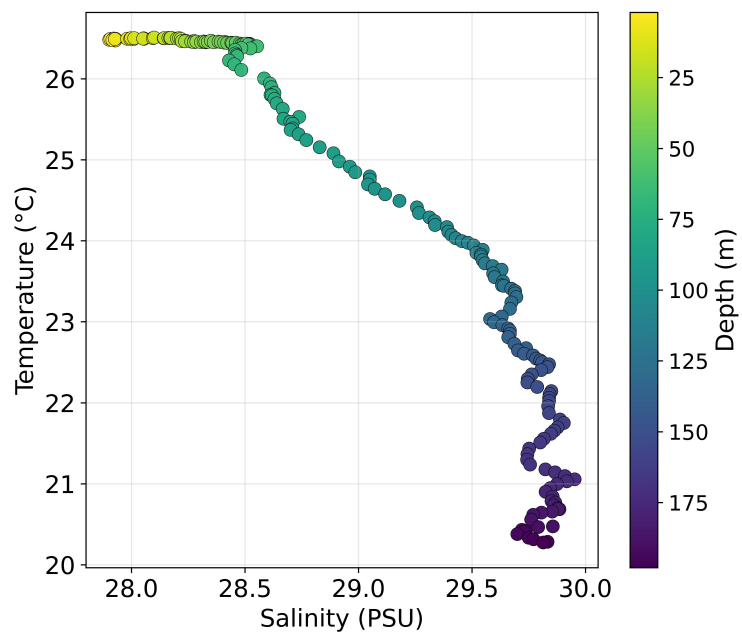


Figure 1.2: Temperature–Salinity (TS) diagram with the same data as in Fig. 1.1

Another important application of CTD data is calculating the speed of sound through the water column, as water temperature and salinity effect the density of seawater, which directly effects sound propagation through the medium. This information is essential for multi-beam sonar mapping of the seafloor, as accurate knowledge of the water's density profile allows for precise correction of sound speed profiles, which directly affects the positioning and depth measurements of the sonar returns.

Variations in water density and temperature can bend or reflect sound waves, impacting areas of naval warfare such as detection range and target location accuracy. Beyond these applications, understanding sound propagation in the water column is critical for underwater communication systems, environmental monitoring, and scientific surveys, where accurate acoustic measurements are necessary for mapping, habitat assessment, and detecting marine life.

1.3. How CTDs are deployed

Generally speaking, a oceanographic research platform can be deployed dynamically or statically. A system can be...

1. Lowered from the surface to vertically profile a column of water, typically called a cast. It could also be affixed to a system that traverses the water column up and down (a winch, wire walker, ocean glider, Argo float, etc.).
2. Affixed to a surface buoy, pier, anchored to the sea floor, or other similarly static location.
3. On a drifting surface float.

Each deployment type has different design considerations, sensor selection, and resulting cost trade-offs. For example, a stationary moored system prioritizes long-term stability, power efficiency, and robustness to harsh conditions, while a mobile profiling float emphasizes compactness, low weight, and energy-efficient data logging. When it comes to sampling the ocean, there are many possible system configurations, ranging from simple single-parameter sensors to fully integrated, multi-sensor platforms capable of measuring temperature, salinity, dissolved oxygen, turbidity, and more. Each configuration presents trade-offs in terms of data resolution, deployment duration, maintenance requirements, and overall cost, meaning the choice of system must align closely with the scientific goals, operational constraints, and environmental conditions of the deployment.

2

SeaLabMOS System Configurations

2.1. Two Modes: Profile and Static

The system has been designed with two operational modes: profile and static. As the system is modular, each mode has different sensors that better suit each mode. When looking at Table 2.1, think about the science questions you are attempting to answer.

1. What do you want to sample (temperature, salinity, location)
2. How deep do you want to sample?
3. Do you want to see live data at the surface or at depth?
4. How long will the system be deployed for (the desired endurance)?

Table 2.1: SeaLabMOS Deployment Types and Platforms

Profile			Static			Live View
			Bottom Pressure Recorder	Temperature Logger	Surface	Wired Profile
1.5 in PVC	2 in PVC	2 in Steel	1.5 in PVC	1.5 in PVC	Nalgene	Pelican case
RP2040	RP2040	RP2040	RP2040	RP2040	RP2040	Raspberry Pi
–	–	–	–	–	GNSS (option)	GNSS
RTC	RTC	RTC	RTC	RTC	RTC/GNSS	GNSS Time
OLED	OLED	OLED	–	–	–	Screen
Switch	Switch	Switch	Switch	Switch	Switch	12 V to 5 V
2.2 Ah	4.4 Ah	4.4 Ah	4.4 Ah	2.2 Ah	4.4, 6.6, 10 Ah	–
–	–	–	–	–	Solar (option)	–
Salt	Salt	Salt	–	–	Salt	Salt
BR Temp	BR Temp	BR Temp	–	–	–	BR Temp
BR Bar30	BR Bar30	BR Bar100	BR Bar02	–	–	BR Bar30
–	Thermistor	Thermistor	–	Thermistor	Thermistor	Thermistor
–	DS18B20	DS18B20	–	DS18B20	DS18B20	DS18B20
–	–	–	–	–	Orange LED	–
140m	140m	350m	140m	140m	2m	140 m–350m

3

Base Hardware Components

Chapter Concepts:

1. What is a microcontroller and how does it work?
2. How to physically connect each component to the microcontroller

3.1. Microcontroller Selection and Base System

The SeaLabMOS main board is the Adafruit Feather RP2040 Adalogger - 8MB Flash with microSD Card. It has an onboard microSD card slot, battery charging capability, and STEMMA QT port for solderless sensor integration. The RP2040 microcontroller has the capability to be overclocked for additional performance and underclocked for additional power savings when endurance is paramount.

Many types/brands of microcontroller could be used, so long as they have an I2C bus and UART capability. Alternatives that may work include:

- Arduino Uno R3 / Adafruit Metro 328. Only fits in larger enclosures and does not have a built in microSD card, but has many shield options. Could be good for bench testing.
- Adafruit QT Py RP2040 (Product ID: Product ID: 4900¹) with Adafruit Lilon or LiPoly Charger BFF Add-On for QT Py (Product ID: 5397²). *Extremely* small with no built in microSD card.
- Teensy 4.1³. Small and powerful. Probably overkill for the current SeaLabMOS use-case unless adapting this work into something more computationally intensive.

The required base parts of the SeaLabMOS are listed in Table 3.1. The OLED display with buttons, GP-S/GNSS, and beacon are mission specific and used when needed, resulting in a cost range:

- \$45 no screen
- \$60 with screen
- \$90 with Mini GPS for location
- \$111.5 with Ultimate GPS for location w/ external antenna

3.2. Base Firmware

The base firmware is the same for all versions of the SeaLabMOS family of systems. The sensors are used by including the relevant header file, as defined on GitHub⁴.

3.2.1. Default Record Format for Comma-Separated-Values

Due to the variable nature of SeaLabMOS, a default header format is used across all firmware. Each firmware type updates only the relevant attached sensor values while leaving the others untouched, and the nth CSV is only populated by the updated values. In this manner, post-processing of data (or live data processing, if building a tethered system) is standardized with one data-ingestion header format.

¹<https://www.adafruit.com/product/4900>

²<https://www.adafruit.com/product/5397>

³<https://www.pjrc.com/store/teensy41.html>

⁴<https://github.com/sscopazzi/SeaLabMOS>

Table 3.1: Base Datalogger Hardware

Name	Adafruit Part #	What it does	Accuracy	Cost
Adafruit Feather RP2040 Adalogger w/ microSD	5980	Datalogger	—	\$15
SD/MicroSD Memory Card (512 MB)	5252	Data storage	—	\$5
DS3231 Precision RTC FeatherWing	3028	Real time clock	±2 minutes per year	\$14
CR1220 12mm 3 V Lithium Coin Cell Battery	380	Battery for RTC	—	\$1
Lithium Ion Battery Pack 3.7 V 2.2 Ah (or 2.5–10 Ah)	1781 (2.2 Ah)	Battery	—	\$10
Adafruit FeatherWing 128×64 OLED display	4650	Screen, Buttons	—	\$15
Mini GPS (PA1010D)	4415	GPS/GNSS, Time	2 m, 10 Hz	\$30
Ultimate GPS Breakout (PA1616D)	5440	GPS/GNSS, Time	meters , 10 Hz	\$30
External Active GPS Antenna	960	For Ultimate	accuracy	\$21.5
High-strength 'rare earth' magnet	9	Hold antenna		\$2.5
STEMMA QT / Qwiic JST SH 4-pin Cable	5385	Sensor Connection		\$1–1.5
Stemma QT / Qwiic 5 Port Hub& 5625	Sensor Connection			\$2.5

Note: Accuracy of RTC from manufacturer. GPS/GNSS module accuracy from testing.

3.2.2. Saving Data to the Onboard microSD Card

All data is indexed by time from the RTC and saved to the onboard microSD card as CSV, with newlines for each record. Dynamic firmware creates a new file at each power-on (e.g., 2025-07-10T14-22-17.csv), while static firmware for multi-day deployments creates or appends one file per day (e.g., 2025-07-21.csv). The datetime format follows ISO 8601[**ISODates**], with “-” replacing “.” to avoid forbidden filename characters.

Profile firmware at 200 MHz (5 Hz) produces 380–450 bytes/s (76–90 bytes/sample), potentially filling a 256 MB card in 7–8 days. Static deployments sample far less frequently, so battery life, not file size, is the limiting factor. Using an 8 GB card removes storage concerns, and smaller cards are feasible if sample duration is altered.

Saving to the onboard microSD restricts data access to post-deployment retrieval. Live monitoring below the surface is not possible, though future wired version(s) of the SeaLabMOS are planned for real-time data monitoring of casts to depth.

3.3. Endurance and Power Monitoring

The endurance is highly variable, mostly due to the battery chosen. To assist in knowing what the battery voltage is upon system power-up, the following table is in the firmware to blink the onboard NeoPixel.

Table 3.2: Battery Voltage Ranges and Corresponding LED Colors

Voltage Range (V)	Color	Behavior	Description
≥ 4.00	Green	Solid	Fully charged
3.80 – 3.99	Cyan	Solid	Mostly full
3.60 – 3.79	Yellow	Solid	Moderate
3.40 – 3.59	Purple	Solid	Low
3.35 – 3.39	Red	Solid	Critically low
< 3.35	Red	Blinking	Dangerously low

3.3.1. Power Use of Each Component

words

3.4. Real-Time Clock

For precise timekeeping, SeaLabMOS uses the DS3231 Precision RTC FeatherWing. The I2C-integrated Real Time Clock (RTC) fits right on top of the base board and has a Temperature Compensated Crystal Oscillator (TCXO) to help maintain accurate timing. The board also has a SQW/INT breakout pin, which is utilized to wake the microcontroller from a low-lower sleep mode.

3.5. OLED Display

If real-time data viewing, in-field pre-deployment button functionality and/or error message monitoring via text is desired, SeaLabMOS can integrate the Adafruit FeatherWing OLED - 128x64 OLED Add-on For Feather⁵. The OLED board also has an extra STEMMA QT connector.

3.6. Location (GPS/GNSS)

If location of the start of the CTD cast is desired, or if building a surface asset, the Adafruit Mini GPS PA1010D⁶ can be integrated through the STEMMA QT port. If this board is integrated into the system and used for location and time, the RTC may not be required. Alternatively, the GNSS board could be utilized for location and time and to set the RTC before entering "sleep mode" (the sensor datasheet terms it Backup Mode) to conserve power.

3.7. Flashing Yellow Lighted Beacon

In the SeaLabMOS underwater configuration an LED beacon is not needed; however, it should be considered for surface assets that will be deployed during hours of darkness and reduced visibility in areas where mariners or others on the water may come across the platform.

Table 3.3: Components Used for Yellow LED Beacon

Component	Specification / Description	Link
Amber (590 nm) Rebel LED on a SinkPAD-II with Side Emitting Optic	74 lm @ 350 mA	luxeonstar.com ⁷
350 mA BuckToot LED Driver – With Leads	Protects the LED	luxeonstar.com ⁸
Adafruit 12 V Bias Voltage Boost Converter – TPS61040	Boosts 3.3 V from microcontroller to 12 V for LED, has Enable pin for on/off	adafruit.com ⁹

The color of the light (590 nm) is important. To maintain international standards for the safety of navigation, the International Association of Marine Aids to Navigation and Lighthouse Authorities (IALA) maintains standards for navigational beacons. The SeaLabMOS, being a scientific device, is within the category of special marks requiring a yellow light. A light must be present during all times of darkness and reduced visibility (fog, smoke, etc.) [source].

3.7.1. Color

As stipulated within *Marine Signal Lights Colours E200-1* from December 2018 [link], IALA strictly denotes the exact color of each light type. All "colours are specified with CIE 1931 standard colorimetric observer (2°-observer)" using Chromaticity Corner Coordinates (CCCs). The document indicates yellow using CCCs as stated in Table 3.4.

Sources will be added later.

Table 3.4: IALA CCCs X and Y vs Nanometer Equivalent

	X	Y	Nanometer Equivalent
IALA Spec	0.5865	0.413	??? (more than 592.2)
Calculated	0.5752	0.424	590.0

The LED selected emits a wavelength of 590 nm, which falls within the yellow light specification required in Annex A, Color Regions

3.7.2. Characteristic (Light Pattern)

As stipulated by IALA in *0110 Rhythmic Characters Of Lights On Marine Aids To Navigation* [link], a special mark may only have certain characteristics. To simplify

⁵<https://www.adafruit.com/product/4650>

⁶<https://www.adafruit.com/product/4415>

1. **Group-occulting:** This setting provides onlookers greater periods of light than darkness for easier identification. The IALA specification for special marks states “the number of eclipses in a group should not be greater than four in general, and should be five only as an exception” [\[link\]](#).
2. **SeaLabMOS group-occulting:** To better meet endurance needs and to help the LED not overheat, the flashing pattern is only shown once per minute (user configurable).

3.7.3. Automatic Timing of the Light

While “time switches, GNSS synchronization, [or] a real-time clock” *can* control light operation, the predominant method for night operation of AtoN lights is via photo-sensitive devices [\[link\]](#). This allows automatic adaptability during poor visibility, such as fog. Care must be taken to select the correct ambient light switching level to minimize energy consumption and material wear. Ambient light levels (in lux) for different conditions are stated in Table ??, and astronomical events with associated lux are listed in Table ?? [\[link\]](#).

In the current state of the SeaLabMOS, there is no light sensor integrated. We are investigating two options that could suit this use-case.

1. Adafruit TSL2591 High Dynamic Range Digital Light Sensor.
2. Adafruit AS7262 6-Channel Visible Light / Color Sensor Breakout.

While the light sensor could be used to control the beacon on a surface asset, we are also investigating the potential for light sensors to simulate PAR sensors.

3.8. Light Sensors

While the light sensor could be used to turn the LED beacon on and off, we are also investigating the potential for use as a low-cost PAR sensor.

- Adafruit TSL2591 High Dynamic Range Digital Light Sensor (Product ID: 1980). Ultra low-light sensor with a wide dynamic range. I2C interface and STEMMA QT.
- Adafruit AS7262 6-Channel Visible Light / Color Sensor Breakout (Product ID: 3779). Provides six discrete wavelength channels (450–950 nm). I2C communication and STEMMA QT.

Research is ongoing. If you have any thoughts, suggestions, ideas, or other thoughts, please reach out!

4

Sensors

4.1. Battery Voltage

Connect a 100K Ω resistor between BAT and A2 and a second 100K Ω from A2 to GND.

1. **VBAT — Analog A2**
2. **Analog A2 — Ground**

Voltage is measured using a resistor voltage divider connected to the Feather's analog input pin as described by Adafruit [adafruit_feather_rp2040]. The Feather's ADC measures 0–3.3 V, so the voltage divider scales the battery voltage down to a range compatible with the ADC, which measures voltages between 0 and 3.3 V. The divider consists of two resistors, R_1 from the battery positive to the analog pin, and R_2 from the analog pin to ground. The voltage at the analog pin is given by:

$$V_{\text{ADC}} = V_{\text{bat}} \cdot \frac{R_2}{R_1 + R_2}$$

where V_{bat} is the battery voltage. The ADC reading is then converted to an actual voltage by scaling the raw value to the 0–3.3 V range of the ADC and then multiplying by the inverse of the voltage divider ratio.

In the firmware, this is implemented as:

```
float v = analogRead(BATTV_PIN); // read ADC
v /= 4095.0;                      // convert to 0-1
v *= 3.3;                        // scale to voltage at analog pin
v *= 2.0;                        // undo 1:1 voltage divider
```

4.1.1. Voltage Correction Factor

Each system built will have a different system-specific correction factor that must be applied to account for resistor tolerances. This factor is determined empirically by comparing the datalogger reading to a multimeter measurement. In the firmware this correction implemented within *batteryMonitor.h* as:

```
#if SYSTEM_NAME == GREEN (for example)
    v *= 1.053;
```

Steps to Find Correction Factor:

1. Ensure the Feather is printing the voltage to the serial monitor
2. Measure from the positive wire of the battery (once you've wired up the power switch this should be easily accessible) to the Feather's ground
3. Update the relevant SYSTEM_NAME's correction factor until the Feather matches the reading

Alternatively, it is also possible to observe the multimeter's voltage reading and record the Feather's voltage reading at the same time. For example, if the multimeter reads $V_{\text{mm}} = 3.53$ V and the Feather reads $V_{\text{Feather}} = 3.67$ V, the correction factor can be calculated as

$$\text{Correction factor} = \frac{V_{\text{Feather}}}{V_{\text{mm}}} = \frac{3.67}{3.53} \approx 1.043.$$

4.2. Salinity: Atlas Scientific EZO-EC Embedded Conductivity Circuit and K1.0 Mini Probe

The SeaLabCTD uses the Atlas Scientific EZO-EC conductivity circuit with the Mini Conductivity K 1.0 kit¹ to measure salinity via electrical conductivity. The EZO-EC circuit supports multiple output formats and is compatible with 3.3 V and 5 V systems, making it ideal for embedded applications. When paired with an accurate temperature sensor, this circuit allows real-time, in-situ salinity calculation using the Practical Salinity Scale (PSS-78).

4.2.1. Wiring

The EZO-EC circuit supports UART and I²C, but the SeaLabMOS uses UART for direct communication with the Feather RP2040 Adalogger wired as follows:

1. **VCC — 3.3 V**
2. **GND — Ground**
3. **TX — RX** (e.g., Serial1 RX)
4. **RX — TX** (e.g., Serial1 TX)
5. **OFF - GPIO Digital Pin D4**

By default on arrival, the EZO-EC should be configured for UART communication. If not, it can be configured for UART by sending the command `I2C,0` over a temporary I²C connection, or by holding the EZO-EC's PGM pad low on power-up for >10 seconds, which resets it to UART mode.

Once wired, it is possible to "talk" with the EZO-EC circuit through the firmware and Arduino IDE. Once the Feather is wired to the sensor as above, it's possible to send the commands listed in the EZO-EC datasheet².

4.2.2. Calibration

The EZO-EC circuit supports one-, two-, or three-point calibration. For highest accuracy across a range of salinity, the SeaLabMOS uses the three-point calibration: dry, low, and high. Before calibrating the probe, set the temperature compensation value to the temperature reported by the Blue Robotics Fast Temp using the command:

```
RT,<value>\r
```

The carriage return ("`\r`") is selected in the Arduino IDE and does not need to be manually typed. This will be used throughout the calibration process, which is as follows:

1. **Dry Calibration:**
 - Remove the probe from all liquids and ensure it is thoroughly dry.
 - Once powered on, allow the probe to stabilize in air away from other electronics for 10–15 seconds.
 - Send `Cal,dry` via UART.
2. **Low-Point Calibration:**
 - If wet and/or exposed or unknown previously, rinse the probe with distilled water and gently blot dry and/or blow dry with compressed air.
 - Type, but DO NOT send, the command that you will send (e.g., `Cal,low,12880`).
 - Immerse the probe in the low-point calibration solution (e.g., 12,880 µS/cm).
 - Wait for readings to stabilize (usually about 5-10 seconds). You'll notice that if left in the calibration solution for too long, the values will slowly drift. Send the calibration command once the value stabilizes, but before it drifts.
 - Send the command `Cal,low,<value>` (e.g., `Cal,low,12880`).
3. **High-Point Calibration:**
 - Rinse again with distilled water and blot the probe dry and/or blow dry with compressed air.
 - Type, but DO NOT send, the command that you will send (e.g., `Cal,low,80000`).
 - Place it in the high-point calibration solution (e.g., 80,000 µS/cm).
 - After readings stabilize, send `Cal,high,<value>` (e.g., `Cal,high,80000`).

¹<https://atlas-scientific.com/kits/mini-conductivity-k-1-0-kit/>

²https://files.atlas-scientific.com/EC_EZO_Datasheet.pdf

Once complete, check the calibration status by sending `Cal,?`. A properly calibrated circuit will return the calibration points currently in use. If you followed the above, it will return 3.

4.2.3. Atlas-Scientific Power Management

To minimize power consumption during long deployments and when not actively sampling, the SeaLabMOS firmware turns the circuit on and off using GPIO Digital Pin 4.

4.3. Temperature and Pressure Sensor Wiring

Find the section that pertains to the sensors you have selected.

4.3.1. Blue Robotics Temp and Pressure

From Blue Robotics, the sensors has a 4-pos JST GH that is compatible with most flight controller boards, including Pixhawk and others (<https://bluerobotics.com/store/sensors-cameras/sensors/bar30-sensor-r1/>) (BM04B-GHS-TBT). However, this connector is useless for our purposes. It must be cut off and replaced with a STEMMA QT / Qwiic JST SH 4-pin Cable.

Use the table below, or check the manufactures websites for which color is connected to what.

Table 4.1: Blue Robotics to Adafruit STEMMA QT cables

Pin / Wire	Blue Robotics	Adafruit STEMMA QT
1	Red – Vin	Red – V
2	Green – SCL	Yellow – SCL
3	White – SDA	Blue – SDA
4	Black – GND	Black – GND

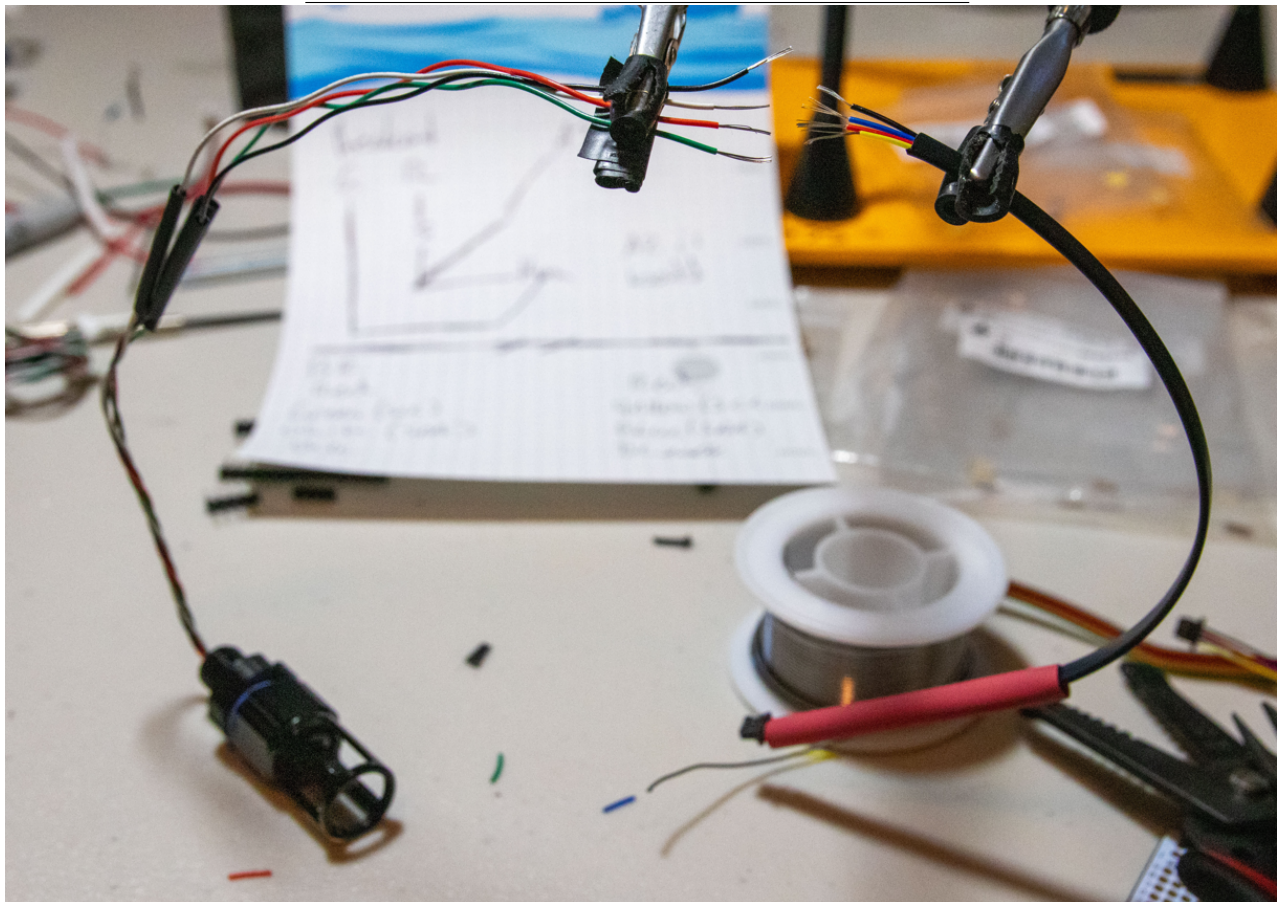


Figure 4.1: Wide photo of Blue Robotics' temperature sensor wiring. Note heat shrink on wires *before* soldering wires together

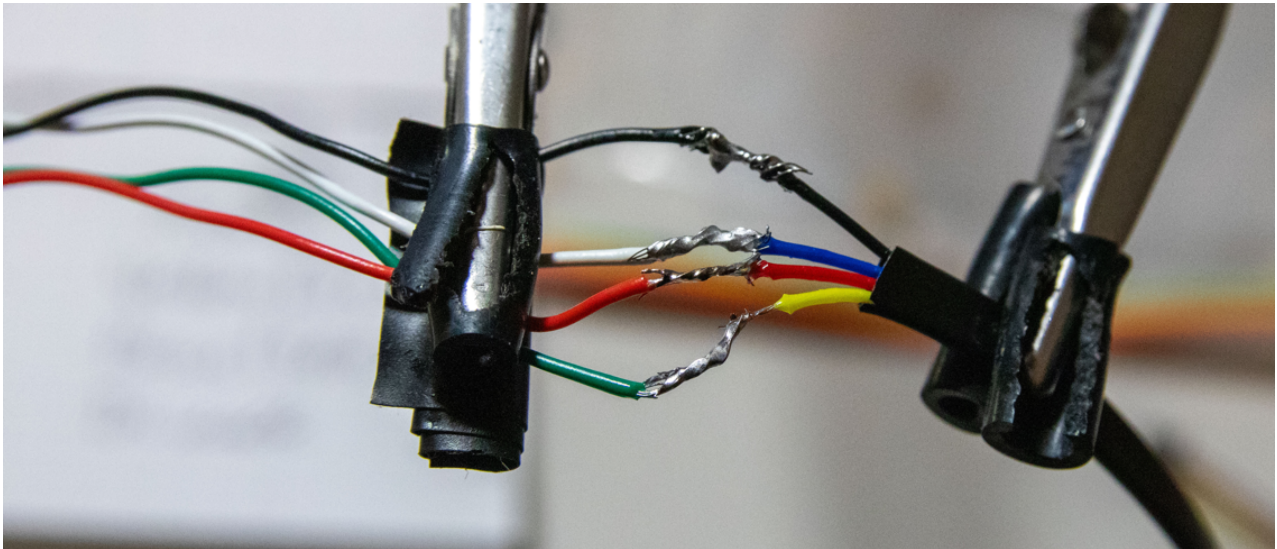


Figure 4.2: Close photo of Blue Robotics' temperature sensor wiring

4.4. Thermistor

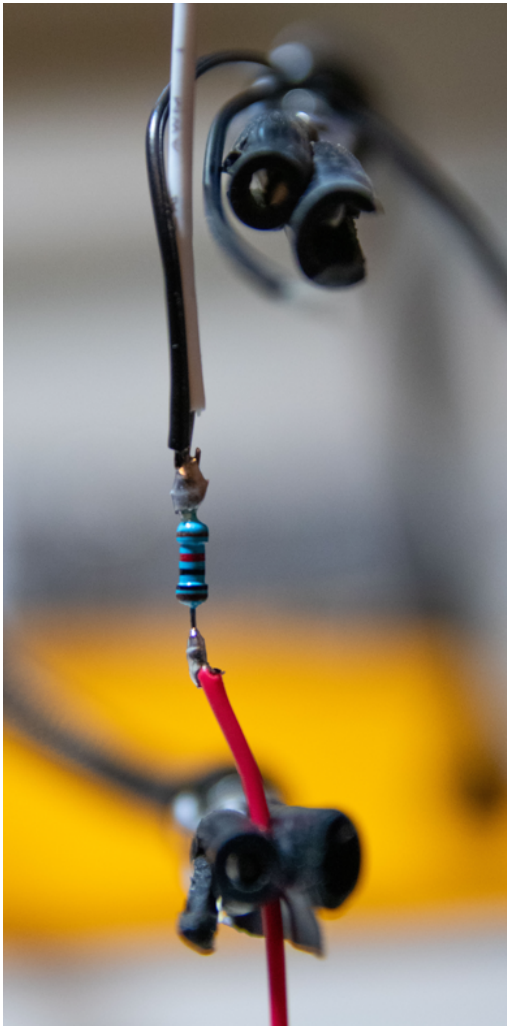


Figure 4.3: Photo of thermistor wiring. White wire goes to Feather's analog port

The thermistor has two wires and comes with a 10 k Ω resistor. The sensor wire (white in the photo) is put on the thermistor side of the resistor that is connected to 3.7V power. Wiring is as follows:

1. **A3** — **Sensor wire (white in photo)**
2. **GND** — **Thermistor wire 1 (either)**
3. **3.3V** — **Resistor** — **Sensor wire and thermistor**

4.4.1. Resistor Accuracy

The thermistor comes with a 10 k Ω resistor used in series; however, it is not perfectly 10 k Ω . To achieve the highest accuracy the resistor should be measured with a multimeter and the value in the firmware updated for the `SYSTEM_NAME` found within the firmware's `therm.h`.

4.4.2. Analog to Digital Resolution

By default, the Feathers analog to digital converter (ADC) resolution is 10-bit with 1024 discrete levels. The Feather board does, however, have the capability for 12-bit resolution with 4095 discrete levels for greater accuracy. Setting the ADC resolution is specified in the firmware's `.ino` file.

5

Default Builds

5.1. 1.5in or 2.0in PVC

Cost How to assemble 3D printed parts and alternatives Endurance

5.2. Temp Logger

Cost How to assemble 3D printed parts and alternatives Endurance

5.3. Bottom Pressure Recorder

Cost How to assemble 3D printed parts and alternatives Endurance

5.4. Steel Pipe

Cost How to assemble 3D printed parts and alternatives Endurance

5.5. Surface Float (Nalgene)